

Question description

the popular engineering college got lowest pass percentage in last semester. the principal conducted faculty meeting and decided to visit all the classes surprisingly.

Dr.Ramprasath is a faculty, who handling data structure course for EEE department second year students.

one day this faculty was handling very interesting topic in data structure such that Linked List,

During this lecture time, principal surprisingly visited to the class and asking to conduct surprise test on Linked list concept.

So the faculty decided to conduct test on the topic of Linked List.

the question was given to last bench students that is,

The nodes are deleted before a certain given node in the linked list.

For example if the given Linked List is 5->10->15->20->25 and

delete before 15 then the Linked List becomes 15->20->25.

Constraint :

$$1 < N < 1000$$

$$1 < P < N-1$$

INPUT Format

First line contains the number of datas- N.

Second line contains N integers(the given linked list).

Question No:7

Third line contains position of the node to be deleted.

OUTPUT Format

Single line represents the final linked list after deletion.

Logical Test Cases

Test Case 1

INPUT (STDIN)

```
4
9 77 12 6
12
```

EXPECTED OUTPUT

```
Linked List:->12->6
```

Test Case 2

INPUT (STDIN)

```
4
91 77 12 61
2
```

EXPECTED OUTPUT

```
Invalid Node! Linked List:->91->77->12->61
```

Mandatory Test Cases

Test Case 1

KEYWORD

```
struct node
```

Test Case 2

KEYWORD

```
node *next;
```

Test Case 3

KEYWORD

```
void create()
```

Code:

```
1  #include <stdio.h>
2  #include <stdlib.h>
3
4  struct Node {
5      int data;
6      struct Node *next;
7  };
8
9  struct Node* createNode(int data) {
10     struct Node *newNode = (struct Node*)malloc(sizeof(struct Node));
11
12     newNode->data = data;
13     newNode->next = NULL;
14
15     return newNode;
16 }
17
18 struct Node* deleteBefore(struct Node *head, int value) {
19     struct Node *temp = head;
20
21     /* Find the node containing value */
22     while (temp != NULL && temp->data != value) {
23         temp = temp->next;
24     }
25
26     /* Node not found */
27     if (temp == NULL) {
28         return NULL;
```

```
29     }
30
31     /* The given node is already the first node */
32     if (temp == head) {
33         return head;
34     }
35
36     /*
37      temp is the required node.
38      Delete all nodes before temp.
39     */
40     struct Node *newHead = temp;
41     struct Node *current = head;
42
43     while (current != temp) {
44         struct Node *deleteNode = current;
45         current = current->next;
46         free(deleteNode);
47     }
48
49     return newHead;
50 }
51
52 void display(struct Node *head) {
```

```
53     struct Node *temp = head;
54
55     printf("Linked List:");
56
57     while (temp != NULL) {
58         printf("-->%d", temp->data);
59         temp = temp->next;
60     }
61 }
62
63 int main() {
64     int n, value;
65
66     scanf("%d", &n);
67
68     struct Node *head = NULL;
69     struct Node *tail = NULL;
70
71     for (int i = 0; i < n; i++) {
72         int data;
73         scanf("%d", &data);
74
75         struct Node *newNode = createNode(data);
76
77         if (head == NULL) {
78             head = newNode;
```

```
79         tail = newNode;
80     }
81     else {
82         tail->next = newNode;
83         tail = newNode;
84     }
85 }
86
87 scanf("%d", &value);
88
89 /* Check whether the node exists */
90 struct Node *check = head;
91
92 while (check != NULL && check->data != value) {
93     check = check->next;
94 }
95 if (check == NULL) {
96     printf("Invalid Node! ");
97     display(head);
98 }
99 else {
100     head = deleteBefore(head, value);
101     display(head);
102 }
103 return 0;
104 }
```

Output:

Test Case-1:

```
4
9 77 12 6
12
Linked List:-->12-->6
```

Test Case-2:

```
4
91 77 12 61
2
Invalid Node! Linked List:-->91-->77-->12-->61
```

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